

**FEASIBILITY STUDY**  
**WAH FOR HOSPITALS (WAH4H)**

**ASIA PACIFIC COLLEGE**  
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**MAKATI CITY, METRO MANILA 1232**

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## 1. Executive Summary

This Feasibility Study evaluates the viability of the WAH for Hospitals (WAH4H) project, namely the containerization, deployment, and feature integration of a Hospital Information System (HIS) onto the Wireless Access for Health (WAH) organization's Proxmox bare-metal infrastructure within a 9.4-week academic MVP schedule.

The WAH4H system, developed during PBL1, exists as an isolated, monolithic Django application with no production hosting, no standardized deployment pipeline, and no compliance with Philippine healthcare regulations. In its current state, the system is operationally inert, inaccessible to WAH stakeholders, government evaluators, or potential institutional partners, and incapable of supporting PhilHealth eClaims processing or PH Core FHIR R4 clinical workflows.

This study evaluates five critical dimensions of feasibility: technical, operational, financial, schedule, and regulatory. Across all five dimensions, the recommended approach (Docker containerization deployed to WAH's existing Proxmox server via Cloudflare Tunnel) is confirmed to be viable. The project's unique financial structure (WAH assumes ₱0 liability for Team Pseudoers' simulated labor; WAH owns all post-closure infrastructure costs) eliminates labor-cost liability for WAH while preserving WAH responsibility for infrastructure planning and operations. The study concludes with a clear recommendation to proceed with full project execution as chartered.

## 2. Description of Products and Services

**Background & Business Need:** The Wireless Access for Health (WAH) organization operates a community health program serving underserved communities in the Philippines. During PBL1, Team Pseudoers developed an initial Hospital Information System covering eight core clinical modules: Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, and Laboratory. While functionally coded, the PBL1 system was built as a localized academic prototype, running only on individual developer machines, with no containerization, no production deployment, and no centralized hosting. This architecture makes the system inaccessible to WAH management, government officials, or academic evaluators without direct physical access to a developer workstation. WAH's strategic objective is to present a demonstration-ready HIS to local government officials (governors and mayors) to secure

institutional buy-in for full-scale deployment. The current system, in its isolated state, cannot fulfill this objective. The WAH4H project directly addresses this gap by transitioning the system from an academic build to a demonstration-ready platform.

**Project Objectives:** Containerize the PBL1 HIS codebase using Docker and deploy to WAH’s Proxmox bare-metal hypervisor via a standardized 3-tier pipeline (Local → Sandbox → Production). Integrate three PBL2 feature modules (Admin, Appointments, and eClaims) alongside security enhancements (JWT authentication, RBAC, and audit logging). Achieve full compliance with Philippine healthcare regulatory standards: PH Core FHIR R4, PhilHealth eClaims, ICD-10-PH, LOINC, SNOMED CT, PSGC, MIMS PH, RVS, RA 10173, and RA 11223. Deliver a complete documentation suite: Training Manual, End-User Manual, Technical Manual, UAT Documentation, and Source Code Repository. Enable WAH to present a fully functional HIS demonstration to government officials and secure institutional commitment for production deployment.

**Scope Summary:** IN SCOPE: PBL1 Modules (Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, Laboratory); PBL2 Modules (Admin, Appointments, eClaims); Cross-System WAH4P/WAH4PC interoperability & data synchronization; Documentation suite. OUT OF SCOPE: Post-delivery maintenance and enhancements after turnover; server hardware procurement (WAH responsibility); PhilHealth accreditation and NHFR registration (WAH prerequisites); external SSL certificates and external cloud infrastructure; ongoing operational support after formal project closure.

### 3. Technology Considerations

The recommended technical architecture leverages industry-proven, open-source technologies directly suited to WAH’s infrastructure constraints.

| Component         | Technology          | Rationale                                      | Feasibility |
|-------------------|---------------------|------------------------------------------------|-------------|
| Backend Framework | Django 5.x (Python) | Mature; PBL1 codebase already built in Django; | HIGH        |

| Component                 | Technology                  | Rationale                                                                     | Feasibility |
|---------------------------|-----------------------------|-------------------------------------------------------------------------------|-------------|
|                           |                             | extensive FHIR library support                                                |             |
| <b>Frontend Framework</b> | ReactJS                     | Existing PBL1 UI; component-based architecture supports PBL2 augmentation     | HIGH        |
| <b>Database</b>           | PostgreSQL                  | Relational integrity required for clinical data; FHIR R4 schema compliance    | HIGH        |
| <b>Containerization</b>   | Docker & Docker Compose     | Environment parity across 4 developer machines; modular PBL2 deployment       | HIGH        |
| <b>Hypervisor</b>         | Proxmox VE (Bare-Metal)     | WAH-owned hardware; eliminates cloud cost; data sovereignty under RA 10173    | HIGH        |
| <b>Secure Tunnel</b>      | Cloudflare Tunnel (\$20/mo) | Existing WAH service; enables external demo access without exposing server IP | HIGH        |
| <b>Security</b>           | JWT + RBAC + Audit Logging  | Required for PH Core FHIR R4 compliance and RA 10173 data privacy obligations | HIGH        |

**Infrastructure Assessment:** WAH currently owns a bare-metal server with the following specifications, confirmed through the meeting with Sir Robert Michael Martinez:

| Resource         | Specification                            |
|------------------|------------------------------------------|
| RAM              | 16 GB                                    |
| Storage          | 320 GB                                   |
| CPUs             | 8 Cores                                  |
| Operating System | Linux                                    |
| Hypervisor       | Proxmox VE (to be installed)             |
| External Access  | Cloudflare Tunnel (existing, \$20/month) |

This hardware is sufficient to host the Dockerized WAH4H system within the project scope. All architecture decisions have been validated against these hardware limits. No infrastructure procurement is required. Server provisioning and physical setup are the responsibility of Sir Robert Michael Martinez (Server/Deployment Manager, WAH), tracked as project assumption A1 and risk R5.

**Deployment Architecture:** 3-tier pipeline designed to eliminate environment inconsistency:

Tier 1: Local Developer Machines (4×): Docker-based development and unit testing; environment parity via pinned base images. Tier 2: Proxmox Sandbox VM: Integration testing and full-system validation; mirrors production configuration. Tier 3: Proxmox Production VM: Demo-ready deployment; Cloudflare Tunnel enabled for external government presentation access. Technical feasibility is confirmed. All required technologies are available, open-source, and already partially in use by the team.

#### 4. Product/Service Marketplace

Team Pseudoers consists of four members with defined roles and complementary competencies. Elijah Josh R. Quibin (Project Manager): schedule tracking, stakeholder communication, documentation control; covers project governance, risk management, all PROJMAN artifacts. Jhon Lloyd Nicolas (Lead Developer / DevOps): Django 5.x, FHIR R4

implementation, Docker, PBL2 security (JWT/RBAC); covers backend architecture, API development, infrastructure configuration. Mariyah Vanna Monique C. Chavez (Quality Assurance): QA validation, PHCore R4 alignment, requirements traceability, UAT oversight, documentation quality review; covers QA validation, UAT oversight, compliance verification. John Kenneth L. Jajurie (Scrum Master / Agile Developer): Agile Scrum facilitation, sprint ceremonies, backlog management; covers sprint facilitation, daily standups, retrospectives.

**Stakeholder Readiness:** Sir Robert Michael Martinez, READY: existing technical capability confirmed; Technical Manual and runbook training included in scope. Ms. Anna Katrina Yturalde, READY: actively engaged; primary point of acceptance; all 3 manuals to be delivered and reviewed before formal sign-off. Sir Robert M. Martinez (server provisioning), DEPENDENT: provisioning must be completed before Milestone 2; PM escalation if server not accessible by April 10, 2026. Ma'am Seann Biag, READY: management oversight confirmed. Sir Pey Canlas, READY: executive sponsor; final sign-off authority confirmed in Charter; PM communicates directly if milestone deadline is at risk. Operational feasibility is confirmed.

## 5. Marketing Strategy

The WAH4H project aligns with the following strategic plans. WAH Digital Health Infrastructure, establish scalable, sovereign, locally managed digital health platform: WAH4H deploys a 100% locally managed HIS on WAH's Proxmox server, eliminating cloud dependency and ensuring data sovereignty. WAH Regulatory Compliance, all 11 modules built to PH Core FHIR R4 standards with PhilHealth eClaims, ICD-10-PH, LOINC, and SNOMED CT integration. WAH Government Partnership, the demo-ready system is the primary vehicle for WAH's government presentations to governors and mayors. WAH Operational Efficiency, WAH4H covers the complete clinical workflow in a single integrated platform, replacing fragmented manual processes. APC Academic Excellence (BSCS-SS231), project managed under a hybrid framework with full documentation artifacts (Charter, BC, SOW, Risk, Stakeholder plans). APC Industry-Relevant Delivery, WAH4H is not a mock project; it is a real system for a real client (WAH) addressing a real healthcare gap in the Philippines.

## 6. Organization and Staffing

The WAH4H project introduces a significant operational shift for WAH, from manual, paper-based clinical workflows to a digitized, integrated HIS. Clinical Workflows, current: manual registration, billing, and discharge documentation; post-WAH4H: digitized end-to-end workflow across 11 modules; transition support: End-User Manual and role-specific training. Staff Training, current: no formal digital health system training; post-WAH4H: all 7 hospital user roles trained via Training Manual; transition support: cascaded internal training enabled by Training Manual. IT Operations, current: no centralized system management; no backup/restore procedures; post-WAH4H: Proxmox-hosted system with documented backup and restore in Technical Manual; transition support: Technical Manual and runbook training for Sir Robert Michael Martinez. PhilHealth Billing, current: manual eClaims filing; post-WAH4H: automated eClaims module (demo-ready; pending WAH accreditation); transition support: eClaims documentation included; go-live contingent on WAH's PhilHealth prerequisites. Data Privacy (RA 10173), current: no formal audit trail or access control; post-WAH4H: JWT + RBAC + audit logging fully implemented and validated; transition support: security compliance checklist verified by QA.

## 7. Schedule

The project operates under a fixed 9.4-week MVP schedule from March 23 to May 27, 2026, driven by the APC BSCS-SS231 academic submission deadline. The schedule is structured as a hybrid predictive-agile framework: high-level milestones are fixed and predictive; technical execution within each milestone uses iterative Agile (Scrum) sprints.

1 – Project Plan, Charter & SOW: Drafted and Approved, 03/24/2026, DONE, FEASIBLE: Completed on schedule. All initiation artifacts approved.

2 – Proxmox Hypervisor Installation & VM Network Bridging, 04/10/2026, DONE, FEASIBLE: Infrastructure dependency met. Sir Robert confirmed server provisioning.

3 – System Architecture & Docker Containerization Design, 04/24/2026, DONE, FEASIBLE: All containers build successfully across developer machines.

4 – Sandbox Integration Testing (100% Environment Parity), 05/08/2026, DONE, FEASIBLE: Environment parity confirmed. Integration tests passing.

5 – PBL2 Feature Integration (Admin, Appointments, eClaims, Security), 05/15/2026, IN PROGRESS, FEASIBLE: Active sprint. On schedule as of May 2026.

6 – Final Deployment: Proxmox Production Initialization, 05/22/2026, PENDING, FEASIBLE: Contingent on Milestone 5 completion. No blocking issues identified.

7 – Client UAT Feedback Received: UAT-requested development changes pending; next UAT/sign-off cycle pending, 05/18/2026, IN PROGRESS, AT RISK: Client UAT feedback received May 18; implementation of requested changes in progress before next UAT/sign-off.

8 – Documentation Completion: All 3 Manuals + UAT Docs, TBD, IN PROGRESS, AT RISK: Parallel documentation work in progress; requires Anna Katrina Yturralde acceptance.

9 – WAH4P/WAH4PC Integration & Data Sync Validation (dependency-gated), TBD, PENDING, AT RISK: Interop spec from Sir Robert Michael Martinez not yet received. Risk R2.

10 – Formal Turnover, Runbook Training & Client Handoff, May–Jun 2026, AT RISK: Dependent on UAT and documentation completion.

11 – Formal Project Closure, Ms. Anna Katrina Yturralde then Sir Pey Canlas final sign-off, May–Jun 2026, AT RISK: Closure contingent on all preceding milestones.

Schedule feasibility is conditionally confirmed. Milestones 1–5 have been completed on or ahead of schedule. Milestones 7–11 carry residual schedule risk actively monitored through weekly sprint reviews, scope freeze effective May 8, 2026, parallel documentation drafting, and direct PM escalation to Sir Pey Canlas if any milestone is projected to slip by more than two weeks.

## 8. Financial Projections

The WAH4H project operates under a unique financial structure: WAH assumes ₱0 liability for Team Pseudoers' simulated labor and owns all post-closure infrastructure costs. Conversion rate: \$1 ≈ ₱56.50 (May 2026).

**Cost Summary:** One-Time Labor: role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr); total planned labor ₱185,158.50, Team Pseudoers liability. Proxmox Server Setup (One-Time): WAH-owned bare-metal server provisioned by Sir Robert



M. Martinez; no hardware procurement required, ₱0 WAH liability. Cloudflare Tunnel (Recurring): Existing service, \$20/month × ₱56.50 × 12 months, ₱13,560.00/yr WAH liability. Cloud Cost Avoided (Savings): Equivalent Linode dedicated server spec (\$144/month) eliminated by Proxmox deployment, ₱97,632.00/yr SAVED.

**Cost-Benefit Comparison:** Status Quo (No Project): ₱0 one-time labor; ₱0 hosting (system unusable); no digitization; non-compliant; no demo capability; ₱0 WAH liability. Cloud Deployment (Linode/AWS): ₱185,158.50 labor; ₱97,632/yr hosting; full digitization; compliant; data sovereignty AT RISK. WAH4H (Recommended): ₱185,158.50 labor (Team; WAH: ₱0); ₱13,560/yr (Cloudflare only); full digitization across 11 modules; compliant; data sovereignty SECURED (local Proxmox); government demo capability: full.

**Return on Investment:** ₱97,632 per year in cloud hosting costs avoided by utilizing WAH's existing Proxmox infrastructure. Estimated 40% reduction in administrative processing time through digitized patient registration, billing, pharmacy, and discharge workflows. Automated PhilHealth eClaims processing reducing billing errors and reimbursement delays (pending WAH's PhilHealth accreditation). Elimination of third-party training vendor costs through cascaded training model enabled by the Training Manual (₱0 external training cost). Institutional buy-in from local government units, the primary strategic prerequisite for WAH securing funding for full production deployment. Financial feasibility is confirmed.

## 9. Findings and Recommendations

**Feasibility Summary:** Technical Feasibility, CONFIRMED: All technologies proven, open-source, and already partially implemented. Hardware confirmed adequate. 3-tier pipeline mitigates deployment risk. Operational Feasibility, CONFIRMED: Team possesses full technical and PM capability. WAH stakeholders are engaged and ready. Complete documentation suite supports independent WAH operation at handover. Financial Feasibility, CONFIRMED: WAH assumes ₱0 liability for Team Pseudoers' simulated labor; WAH owns all post-closure infrastructure costs. Annual hosting cost is ₱13,560 (Cloudflare only) vs. ₱97,632 for equivalent cloud hosting, a net saving of ₱97,632/year. Schedule Feasibility, CONDITIONALLY CONFIRMED: Milestones 1–5 completed on schedule. Milestones 7–11 carry residual risk. Active mitigation controls in place. Escalation protocols defined. Regulatory Feasibility, CONFIRMED: All 10 applicable Philippine healthcare

standards addressed within project scope. WAH's two external prerequisites (PhilHealth, NHFR) are documented and out of scope.

**Alternatives Analysis:** Status Quo (No Project), NOT SELECTED: system operationally inert; zero value; no government demonstration possible; fails all project objectives. Managed Cloud Deployment (Linode/AWS), NOT SELECTED: violates zero-cost constraint (₱97,632/yr recurring); creates data sovereignty risk under RA 10173; external vendor dependency. Docker on New Hardware, NOT SELECTED: capital expenditure violates ₱0 client liability constraint; procurement timeline incompatible with 9.4-week schedule; WAH already owns suitable hardware. Monolithic Deployment Without Docker, NOT SELECTED: creates deployment inconsistency; eliminates modular PBL2 integration; no rollback capability; fails environment parity requirement. RECOMMENDED, Docker on WAH Proxmox: SELECTED, ₱0 recurring infrastructure cost; data sovereignty; environment parity; modular PBL2 integration; Cloudflare security; bare-metal competency demonstrated.

**Recommendation:** Based on the analysis presented across all five feasibility dimensions, Team Pseudoers formally recommends immediate approval and full execution of the WAH for Hospitals (WAH4H) Hospital Information System project as described in the Project Charter (v1.2) and Business Case (v1.1). The project is technically sound, operationally supported, financially viable, schedule-managed, and regulation-compliant. The recommended approach delivers the maximum organizational benefit at the minimum financial cost and risk to WAH. The single conditional finding (schedule feasibility for Milestones 7–11) is actively managed and does not constitute a blocker to project authorization. Key Decision Factors: WAH owns the necessary hardware, with no capital expenditure required; Team Pseudoers possesses the technical capability to execute all deliverables within the 9.4-week schedule; the project is the only pathway to WAH achieving its government demonstration objective; all 10 applicable Philippine healthcare regulatory standards are addressed within project scope; WAH assumes ₱0 liability for Team Pseudoers' simulated labor, with ₱97,632/year in hosting cost savings versus the cloud alternative.

Approval/signature collection for this section is consolidated in Section 9, Approvals and Authorization of the master project document.